OpenRadioss Workbench for FreeCAD: LS-DYNA Workflow Documentation

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Abstract

This document provides comprehensive documentation for the OpenRadioss workbench, a FreeCAD extension that enables seamless creation and management of LS-DYNA input files (.k files). The workbench provides an integrated environment for finite element analysis preprocessing, combining FreeCAD's powerful 3D modeling capabilities with LS-DYNA's simulation power.

Contents

1 Introduction **2**
- 1.1 Overview . . . 2
- 1.2 Motivation . . . 2
- 1.3 Key Features . . . 2

2 Architecture and Design **2**
- 2.1 Organizational Structure . . . 2
- 2.2 Component Structure . . . 3
  - 2.2.1 GUI Components . . . 3
  - 2.2.2 Command Components . . . 3
  - 2.2.3 Core Logic . . . 3

3 User Workflow **3**
- 3.1 Getting Started . . . 3
  - 3.1.1 Installation . . . 3
  - 3.1.2 Initial Setup . . . 4
- 3.2 Creating LS-DYNA Files . . . 4
  - 3.2.1 Step 1: Create Basic Template . . . 4
  - 3.2.2 Step 2: Edit Content . . . 4
  - 3.2.3 Step 3: Export to .k File . . . 4
- 3.3 Advanced Workflow . . . 5
  - 3.3.1 Using the Keyword Editor . . . 5
  - 3.3.2 Project Organization . . . 5

4 Technical Implementation **5**
4.1 Text Document Management 5
4.1.1 Creating Text Documents 5
4.1.2 Content Structure 5
4.2 Export System 6
4.2.1 Export Process 6
4.2.2 Error Handling 6
4.3 GUI Integration 6
4.3.1 Workbench Registration 6
4.3.2 Command Pattern 7

5 LS-DYNA Template Structure **7**
5.1 Control Cards 7
5.2 Database Output 7
5.3 Material and Section Definitions 7
5.4 Geometry Definitions 8

6 File Format and Persistence **8**
6.1 FreeCAD Document Storage 8
6.2 Export Format 8

7 Integration with LS-DYNA **9**
7.1 Compatibility 9
7.2 Best Practices 9

8 Development and Extension **9**
8.1 Code Organization 9
8.2 Adding New Keywords 9
8.3 Customization 10

9 Troubleshooting **10**
9.1 Common Issues 10
9.1.1 Export Failures 10
9.1.2 GUI Issues 10
9.2 Debug Information 10

10 Conclusion **10**
10.1 Future Enhancements 10
10.2 Contributing 11

1 Introduction

1.1 Overview

The OpenRadioss workbench is a specialized FreeCAD extension designed to bridge the gap between 3D modeling and finite element analysis. It provides tools for creating, editing, and exporting LS-DYNA input files directly within the FreeCAD environment.

1.2 Motivation

Traditional finite element analysis workflows require multiple software tools:

- CAD software for geometry creation
- Preprocessing software for meshing and analysis setup
- Solver input file generation
- Post-processing and visualization

The OpenRadioss workbench consolidates these steps by providing an integrated solution within FreeCAD.

1.3 Key Features

- Interactive LS-DYNA keyword editor with parameter management
- Template-based .k file generation with predefined structures
- Text-based document objects for .k file storage and editing
- Export functionality for generating production-ready .k files
- Persistent storage in FreeCAD's native .FCStd format

2 Architecture and Design

2.1 Organizational Structure

The OpenRadioss workbench follows a hierarchical structure that organizes functionality into logical layers. The following organizational chart provides a visual representation of the system architecture:

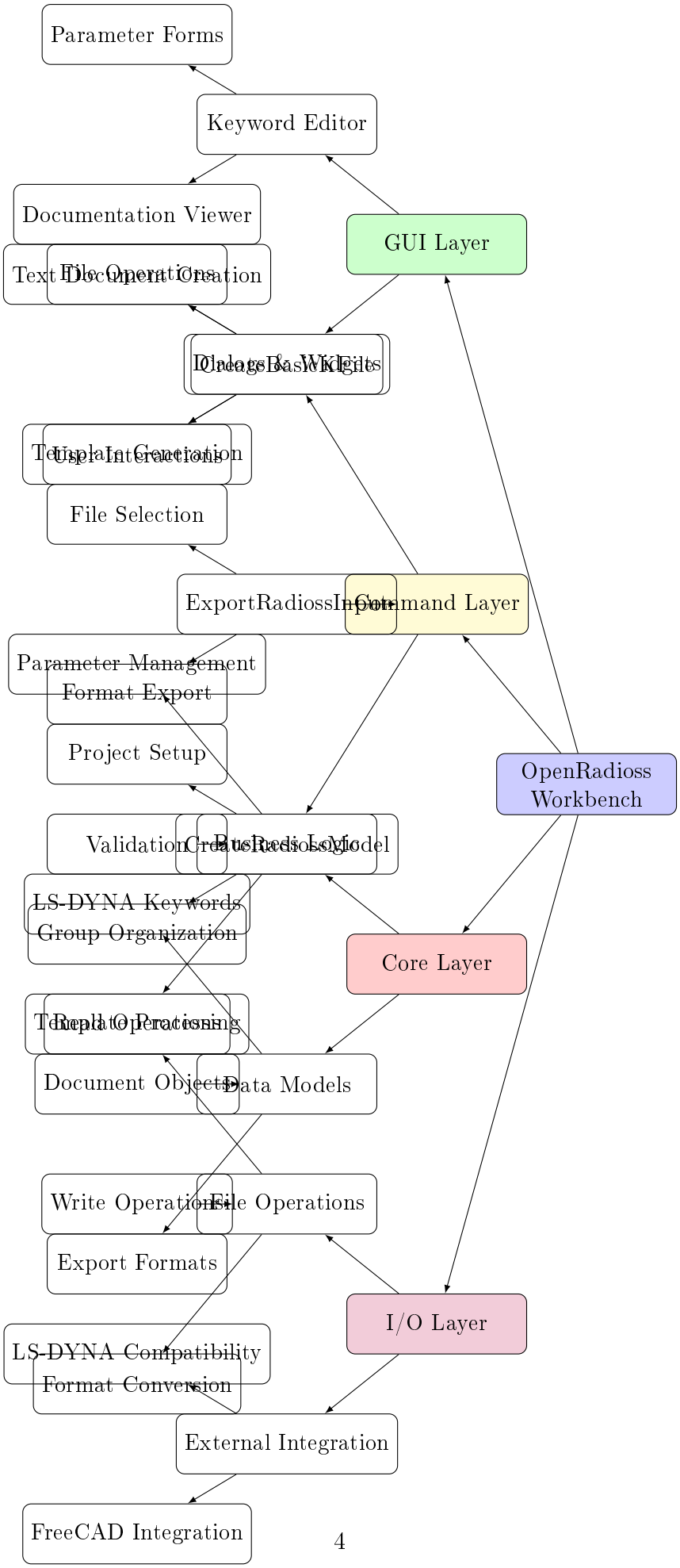


Figure 1: OpenRadioss Workbench Organizational Structure (Rotated 90° CCW)

2.2 Architecture Layer Details

This section provides detailed explanations of each architectural layer shown in the organizational chart.

2.3 Component Structure

2.3.1 GUI Components

- `KeywordEditorDialog`: Main interface for keyword parameter editing
- `DocumentationViewer`: Integrated help and documentation system
- Custom dialogs for file operations and user interactions

2.3.2 Command Components

- `CreateBasicKFile`: Creates LS-DYNA text document objects
- `ExportRadiossInput`: Exports .k files from text documents
- `CreateRadiossModel`: Initializes new analysis projects

2.3.3 Core Logic

- Parameter parsing and validation
- LS-DYNA keyword template management
- Text document object management
- Export formatting and validation

3 User Workflow

3.1 Getting Started

3.1.1 Installation

The OpenRadioss workbench is installed as a standard FreeCAD workbench:

1. Copy the workbench directory to FreeCAD's Mod directory
2. Restart FreeCAD
3. Activate the workbench from the workbench selector

3.1.2 Initial Setup

1. Launch FreeCAD
2. Select "OpenRadioss" from the workbench dropdown
3. The workbench toolbar and menus will be available

3.2 Creating LS-DYNA Files

3.2.1 Step 1: Create Basic Template

Click "Create Basic .k File" to generate a new LS-DYNA text document:

```
# Creates App::TextDocument object with LS-DYNA content
k_file_obj = doc.addObject("App::TextDocument", "LSDynaKeywords")
k_file_obj.Text = generate_basic_k_file_content()
```

Listing 1: Creating LS-DYNA text document

The generated template includes:

- Control cards (termination, timestep, output)
- Database output definitions
- Material and section definitions
- Sample geometry (nodes and elements)

3.2.2 Step 2: Edit Content

The .k file content can be edited in several ways:

1. **Property Panel**: Select the object and edit the "Text" property
2. **Export and Edit**: Export to .k file, edit externally, then re-import
3. **Keyword Editor**: Use the advanced keyword editor for parameter management

3.2.3 Step 3: Export to .k File

Click "Export LS-DYNA .k File" to generate the final .k file:

```
# Find TextDocument objects
for obj in App.ActiveDocument.Objects:
    if obj.TypeId == 'App::TextDocument' and obj.Text:
        # Export obj.Text to .k file
        with open(filepath, 'w') as f:
            f.write(obj.Text)
```

Listing 2: Export functionality

3.3 Advanced Workflow

3.3.1 Using the Keyword Editor

The keyword editor provides advanced parameter management:

1. Select keywords from categorized lists
2. Configure parameters through GUI forms
3. Generate keyword text with parameter substitution
4. Preview and validate before export

3.3.2 Project Organization

- Create separate FreeCAD documents for different analyses
- Use document groups to organize related objects
- Maintain version control through FreeCAD's document system

4 Technical Implementation

4.1 Text Document Management

4.1.1 Creating Text Documents

```
def create_k_file_document(self):
    """Create LS-DYNA text document in FreeCAD."""
    doc = App.ActiveDocument or App.newDocument("LSDyna_Model")

    # Create TextDocument object
    k_file_obj = doc.addObject("App::TextDocument", "LSDynaKeywords")
    k_file_obj.Label = "LS-DYNA Keywords"
    k_file_obj.Text = self._generate_basic_k_file_content()

    return k_file_obj
```

Listing 3: Text document creation

4.1.2 Content Structure

The .k file content follows LS-DYNA format specifications:

- `*KEYWORD`: File header
- `*CONTROL_*`: Simulation control parameters
- `*DATABASE_*`: Output specification
- `*PART`: Part definitions
- `*SECTION_*`: Section properties
- `*MAT_*`: Material definitions
- `*NODE`: Nodal coordinates
- `*ELEMENT_*`: Element connectivity
- `*END`: File termination

4.2 Export System

4.2.1 Export Process

1. Identify TextDocument objects in active document
2. Extract text content from objects
3. Present selection dialog if multiple objects exist
4. Write content to .k file with proper formatting
5. Open file in external editor if available

4.2.2 Error Handling

```
try:
    # Export logic
    success = export_k_file(selected_obj, filepath)
    if success:
        show_success_message(filepath)
except Exception as e:
    show_error_message(f"Export failed: {str(e)}")
```

Listing 4: Export error handling

4.3 GUI Integration

4.3.1 Workbench Registration

```
class OpenRadiossWorkbench(Workbench):
    def Initialize(self):
        # Import and register commands
        from .commands.create_basic_k_file import CreateBasicKFile
        from .commands.export_radioss_input import ExportRadiossInput

        Gui.addCommand('CreateBasicKFile', CreateBasicKFile())
        Gui.addCommand('ExportRadiossInput', ExportRadiossInput())

        # Create toolbar and menu
        self.appendToolbar("OpenRadioss", [
            "CreateBasicKFile", "ExportRadiossInput"
        ])
```

Listing 5: Workbench initialization

4.3.2 Command Pattern

Each workbench command follows the standard FreeCAD command pattern:

```
class ExampleCommand:
    def GetResources(self):
        return {
            'Pixmap': 'icon_path',
            'MenuText': 'Command Name',
            'ToolTip': 'Description'
        }
```

```

    def Activated(self):
        # Command implementation
        pass

    def IsActive(self):
        # Availability check
        return True
```

Listing 6: Command class structure

5 LS-DYNA Template Structure

5.1 Control Cards

```
*CONTROL_TERMINATION
$   endtim    endcyc     dtmin     endeng    endmas
      1.0
$
*CONTROL_TIMESTEP
$   dtinit    scft      isdo      tsd0      dt2ms      dtmp      dt2ms
      0.0     0.0      0.0      0.0
```

Listing 7: Control card generation

5.2 Database Output

```
*DATABASE_BINARY_D3PLOT
$   dt/cycl   lcdt      ioopt
    0.01
$
*DATABASE_GLSTAT
$   dt/cycl   lcdt      ioopt
    0.01
```

Listing 8: Database output configuration

5.3 Material and Section Definitions

```
*PART
$      pid     secid       mid     eosid      hgid      grav     adpopt
         1         1         1         0         0         0         0
$
*SECTION_SHELL
$      sid    elform      shrf       nip     propt   qr/irid     icomp
    setyp
         1         2     1.000     3.000         0         0         0
         0
$
*MAT_ELASTIC
$      mid        ro         e        pr        da        db         k
         1    7.8e-9     2.0e5      0.30
```

Listing 9: Material definitions

5.4 Geometry Definitions

```
*NODE
$      nid                x               y               z       tc
    rc
           1        0.000000        0.000000        0.000000        0        0
           2        1.000000        0.000000        0.000000        0        0
           3        1.000000        1.000000        0.000000        0        0
           4        0.000000        1.000000        0.000000        0        0
$
*ELEMENT_SHELL
$      eid     pid       n1       n2       n3       n4
           1        1        1        2        3        4
```

Listing 10: Sample geometry

6 File Format and Persistence

6.1 FreeCAD Document Storage

The LS-DYNA content is stored as TextDocument objects within FreeCAD's native .FCStd format:

- Document objects persist across FreeCAD sessions
- Content is stored in compressed binary format
- Version control compatible through FreeCAD's document system
- Automatic backup and recovery support

6.2 Export Format

The .k file export maintains exact LS-DYNA formatting:

- Proper keyword ordering and structure
- Correct column alignment for numerical data
- Comment preservation
- LS-DYNA compatibility verification

7 Integration with LS-DYNA

7.1 Compatibility

The generated .k files are compatible with:

- LS-DYNA R7 and later versions
- Standard LS-DYNA ASCII input format
- Common LS-DYNA keywords and parameters

7.2 Best Practices

1. Validate generated .k files before production use
2. Use appropriate units (consistent unit system)
3. Verify material properties and boundary conditions
4. Test with simple models before complex analyses

8 Development and Extension

8.1 Code Organization

```
freecad/ORfemn/
          __init__.py           # Main workbench class
          init_gui.py           # GUI initialization
          gui/                  # GUI components
                keyword_editor.py # Advanced keyword editor
          commands/             # Command implementations
                create_basic_k_file.py
                export_radioss_input.py
                create_radioss_model.py
          docs/                 # Documentation
```

Listing 11: Project structure

8.2 Adding New Keywords

To add support for new LS-DYNA keywords:

1. Update the keywords JSON database
2. Add parameter definitions and validation
3. Implement GUI forms for parameter input
4. Add export formatting logic

8.3 Customization

The workbench can be customized by:

- Modifying template content in `_generate_basic_k_file_content()`
- Adding new command classes for specialized workflows
- Extending the keyword database with domain-specific keywords

9 Troubleshooting

9.1 Common Issues

9.1.1 Export Failures

- Check file permissions and disk space
- Verify TextDocument object exists and has content
- Ensure proper file path specification

9.1.2 GUI Issues

- Restart FreeCAD if workbench doesn't load
- Check Python path and module imports
- Verify Qt/PySide installation compatibility

9.2 Debug Information

Enable debug logging by setting:

```
import logging
logging.basicConfig(level=logging.DEBUG)
```

10 Conclusion

The OpenRadioss workbench provides a powerful and intuitive solution for LS-DYNA preprocessing within FreeCAD. By combining the strengths of both platforms, it offers a streamlined workflow from 3D modeling to finite element analysis.

10.1 Future Enhancements

- Advanced meshing integration
- Post-processing result visualization
- Multi-physics coupling support
- Cloud-based collaboration features

10.2 Contributing

Contributions are welcome through:

- Bug reports and feature requests
- Code contributions and improvements
- Documentation enhancements
- Test case development